5. Save your model configuration.

Deploy a Model

Context

When you have created a model, you can deploy the model directly.

Procedure

1. Launch the MM app.
2. On the **Model Overview** page, click the model you want to deploy. The detail page of the model then appears.
3. On the **Model Details** page, click the **Deploy** button on the top right. The model is then deployed.

Model Header

Context

In the Header tab, you can fill in the following basic information about the model:

- Name
- Description
- Namespace
- Correlation Level.

Correlation level:

- is the level of event correlation propagation between tracked processes within a model.
- is specified as an integer from 1 to 5.

Each tracked process has a list of observed tracked processes. You can add a tracked process to the observed tracked process list by the tracked process creation/update event or by some other actual events.

For example, suppose there is an event correlation chain between tracked processes A, B, and C, which means that tracked process A observes tracked process B and tracked process B observes tracked process C.

- If the model is configured to have 1 level of correlation, then events reported to tracked process C are also correlated to B.
- If the model is configured to have 2 levels of correlation, then events reported to tracked process C are also correlated to both A and B.
- And so on.



Note that the event correlation chain between tracked processes breaks up in the following two cases:

- When tracked processes are of the same type:

  In this example, if tracked processes A and C are of the same type, then the event correlation chain breaks up between B and A. So events reported to tracked process C are correlated to B, but not to A.

- When the actual event's event type is not defined as an admissible planned or unplanned event in this tracked process:

  In this example, if a Proof of Delivery planned/unplanned event has been defined in tracked processes C and A, but not in B, then when the Proof of Delivery actual event is reported to C, the event correlation chain breaks up between B and C. So the Proof of Delivery actual event reported to C won't be correlated to B and A, even if there is a Proof of Delivery planned/unplanned event defined in A.

Also note that the authorization for instances can also be inherited and checked through correlation. So in this example:

- If you have authorization for tracked process C, you will also have authorization for B and A. Then events reported to C are also correlated to B and A.
- If you don't have authorization for tracked process C, the system will check if you have authorization for B. If so, events will be reported to B and correlated to A. Otherwise, the system will check if you have authorization for A.
- If tracked process C doesn't exist, the system will check if you have authorization for B. If so, events will be reported to B and correlated to A. Otherwise, the system will check if you have authorization for A.

Procedure

To complete the model header, do the following:

1. Fill in the mandatory field: **Name**. It must meet the following requirements:
   - It's unique from other models in the same tenant.
   - It starts with a lowercase letter and then lowercase letters or numbers.
   - It cannot start with letters "gtt".
   - It's less than 50 characters.
2. When the model name is set, the **Namespace** will be filled automatically.
3. Fill in the **Description**.
4. Fill in the **Correlation Level** in the range from 1 to 5.

Tracked Process

A tracked process is the main object in the model. Each model must have at least one tracked process. The name of the tracked process must be unique in the model. You can create/edit/delete your own tracked process. You can have at most 200 tracked processes in a model.

In the **Tracked Process** tab, user-defined tracked processes are listed on the left. Select one tracked process in the list. On the right, there are different views of the tracked process:

- **Field Lists**
- **Admissible Planned Events**
- **Admissible Unplanned Events**.

Related Information

Basic Terms

Create Tracked Process

Context

You can create a tracked process in the **Tracked Process** tab.

Procedure

1. In the **Tracked Process** tab, click the **Create** button above the list.
2. Fill in the following:
    - **Name**: mandatory and must be unique in this model.
    - **Description**
    - **Tracking ID Type**: mandatory and must be unique across models in the whole tenant.
    - **Affect Planned Event Status By**: mandatory. You can select either the **Actual Business Timestamp** or **Actual Technical Timestamp** in the drop-down list. The first one is the default. Your selection decides the timestamp the system uses to process the last reported actual event. In turn, this affects the planned event status. For more, see Timestamps.
    - **Sort Planned Events By**: mandatory and you can select either **First Planned Business Timestamp then Payload Sequence** or **First Payload Sequence then Planned Business Timestamp** in the drop-down list. The first one is the default. Your selection decides the sequence of planned events that is used for the tracked process status calculation. You can see the displayed sequence in the **Planned Events** tab of the CPE app.
3. Click **OK**.

Edit Tracked Process

Context

After a tracked process is created, you can edit the tracked process attributes.

Procedure

1. In the **Tracked Process** tab, click the tracked process that you want to change.
2. Click the **Edit** button above the list.
3. In the pop-up dialog, you can edit its name, description and tracking ID type, and change your selection in **Affect Planned Event Status By** and **Sort Planned Events By**.
4. When your editing is complete, click **OK**. The **Model Details** page is changed to the edit mode.
5. Save your model configuration.

Delete Tracked Process

Context

You can delete a tracked process.

Procedure

1. In the **Tracked Process** tab, click the tracked process that you want to delete.
2. Click the **Delete** button above the list.
3. There are two situations:
    - If this tracked process is not used by other objects, before deletion, a warning message box pops up to ask you if you want to delete the tracked process, its fields, admissible planned events, and admissible unplanned events.
    - Otherwise, a **Where Used List** pops up to ask you if you want to delete the tracked process and all the data in the places where this tracked process is used.
4. If you click the **OK** button, the deletion takes effect immediately and the **Model Details** page is changed to the edit mode.
5. Save your model configuration.

Field Lists

In the **Tracked Process** tab, by default, the content of **Field Lists** is displayed with two tables:

- **User Model Fields**: This table contains the user-defined fields in this tracked process. You can create/edit/delete these fields for this tracked process.
- **Core Model Fields**: This table contains all the fields from the predefined core model's process types. You cannot modify them.

Besides, in a standard tracked processes of the standard model "gttft1", you can also see a **Standard Model Fields** table. It contains the standard fields in this tracked process. You can edit a string-type or code list-type standard field to add a role attribute.

You can have at most 200 fields in a tracked process.

Edit Standard Model Fields

Context

To edit a standard model field, do the following:

Procedure

1. Click the pencil icon at the end of the field that needs to be changed in the **Standard Model Fields** table.
2. Edit in the pop-up dialog. You can only do the following changes:
    - Change the visibility attribute of the field:

      If you set a field as **Visible**, it's available in the following UI interfaces of this app, otherwise, it's not:

        - Filters and the worklist settings
        - **Information** tab of the detail pages.

      This complements the adapting UI functionality for hiding fields.

      > **Note**
      > While most fields can be hidden using your model configuration, some types of fields will always remain visible in fulfillment tracking apps. These include:
      >
      > - Fields Integrated with Master Data: Fields linking to master data (for example, location or business partner master data) must be shown.
      > - Combined Fields: fields that combine data from source (for example, **Departure Location / Departure Time**). These are considered essential for the core functionality and data integrity of the tracking applications and cannot be hidden.
      > - Code List Fields with Colors or Icons: Fields that use visual indicators for status (for example, **Delay Status**, **Transportation Mode**).
      > - Location-Relevant Timestamp fields: Timestamps tied to a geographical location (for example, **Actual Arrival At**).

    - Assign role attribute: You can edit a string-type or code list-type standard field to add a role attribute. For more information on role attributes, see Specify Attributes in the Role Using “ReadServiceAttributeTemplate”
3. Click **OK** to save changes. The **Model Details** page is changed to the edit mode.
4. Save your model configuration.

Create User Model Field

Context

To create a user model field, do the following:

Procedure

1. Click the **Create** button above the **User Model Fields** table.
2. In the pop-up dialog, fill in the following information:

Field	Explanation
Name*	The name must be unique in one tracked process.
Label	

Field	Explanation
Type*	You can select one of the following types: • String: You must also specify the length. Enter a number from 1 to 4000. • UUID • Boolean • Integer • Decimal: You must also specify the precision and scale. For Precision, enter a number from 1 to 34. • Date • Timestamp • Association to One: associates to a tracked process, or linked to a tracked process. If you choose this type, you must select a process type. • Association to Many: associates to more than one tracked process, based on an Association to One field of this tracked process. If you choose this type, you must select a process type and one of its Association to One field. You can also input values freely to create new fields directly. • Composition: an item-level table with a structure. If you choose this type, you must select a field type that you defined in the Field Type Pool tab. • Code List: a code list with the string type and a value range. If you choose this type, you must select a code list that you defined in the Code List tab. • Attachments: ▪ Supported file types are: jpeg, jpg, gif, png, pdf, doc, docx, xls, and xlsx. ▪ The maximum file size limit is 10 MB. ▪ Attachments-type fields don't support IDOC integration mapping. ▪ Only the event's attachments- type fields support visibility provider integration mapping.
DPP	The abbreviation for data protection and privacy. It includes the following selectable values: Personal Identifiable Information and Data Subject ID. • There must be only one "Data Subject ID" in one tracked process. • Fields annotated as "Data Subject ID" include the properties of "Personal Identifiable Information". • Changes to the fields annotated as "Data Subject ID" trigger updates of the PDM data visibility, Audit Log Viewer read log and Audit Log Viewer change log.

Field	Explanation
	○ Changes to the fields annotated as "Personal Identifiable Information" trigger updates of the PDM data visibility and Audit Log Viewer change log. ○ For more information, refer to the chapters Security: General and Security: Data Protection and Privacy in the *Administration Guide for Solution Owners*.
Grant	Only string-type fields can be granted **Read** or **Report**. ○ **Read**: If this field is granted "Read" and the user's LBN ID is equal to this field, then you can read this tracked process where the field exists. ○ **Report**: If this field is granted "Report" and the user's LBN ID is equal to this field, then you can write this tracked process where the field exists.
Role Attribute	A role attribute can only be specified to a string-type or code list-type field. It's used to check if a user role has authorization to read this tracked process based on the value of the role attribute defined in SAP BTP cockpit. ○ You can assign at most nine attributes, from `gttUserAuthAttribute001` to `gttUserAuthAttribute009`, to nine different fields in a tracked process. ○ To use this function, you must add the role `ReadServiceAttributeTemplate` to your role collection and specify attributes in this role in SAP BTP cockpit. For more, see the section Specify Attributes in the Role Using "ReadServiceAttributeTemplate" in the *Administration Guide for Solution Owners*.
Readable	When reading the tracked process, this field is readable by the OData service.
Writable	When writing the tracked process, this field is writable.
Visible	This attribute controls the visibility of the fields in fulfillment tracking apps: ○ If the box is checked, the field is displayed in the fulfillment tracking apps by default. You can hide it with the Adapt UI. ○ If the box is unchecked, the field can't be displayed in the fulfillment tracking apps, even with the Adapt UI. A field must be readable in order to be visible.

* mandatory field

3. Click **OK** to create this field.
4. Save your model configuration.

Edit User Model Field

Context

To edit a user model field, do the following:

Procedure

1. Click the pencil icon at the end of the field that needs to be changed in the table.
2. Edit in the pop-up dialog.
3. Click **OK** to save changes. The **Model Details** page is changed to the edit mode.
4. Save your model configuration.

Delete User Model Field

Context

To delete one or more user model field, do the following:

> **Note**
> In the fulfillment tracking apps, after you add a field that is created in the standard model "gttft1" with Adapt UI, deleting it or marking it unreadable in the standard model will lead to errors and improper display of this field.

Procedure

1. Select the fields that you want to delete. Multiple selection is supported.
2. Click the **Delete** button above the **User Model Fields** table.
3. The **To Be Deleted Objects** dialog pops up to ask you if you want to delete the selected fields and all the data in the places where these fields are used.
4. Click **OK**. Fields you selected are removed immediately, and the **Model Details** page is changed to the edit mode.
5. Save your model configuration.

Add URL for User Model Field

Context

To add a URL for a user model field, do the following:

Procedure

1. On the **Model Details** page, click the **Edit** button on the top right.
2. Select a user model field whose type is not "Composition", "Attachments", or "Association to Many".
3. Click the **Add URL** button above the **User Model Fields** table.
4. Fill in the name and address. The address supports variables. Variables should be in curly brackets, such as `{Field1}`. The following fields can be used as variables:

- A user model field whose type is not "Composition", "Attachments", or "Association to Many" in this tracked process.
- A user model field whose type is not "Composition", "Attachments", or "Association to Many" in the associated tracked process, if the type of the field that the URL is added for is "Associate to One".

Note

If you want to navigate to the external systems from the fulfillment tracking apps, you must add URLs for user model fields in the standard model “gttft1”.

You can add either of the following two types of address:

- Address with a fixed URL prefix. For example: ***www.sap.com/?zzdeliveryNo={variable}***
- Address with a URL prefix variable. For example: ***{URL prefix variable}/?zzdeliveryNo={variable}***

Delete URL for User Model Field

Context

To delete a URL for a user model field, do the following:

Procedure

1. On the **Model Details** page, click the **Edit** button on the top right.
2. Find the field whose URL you want to delete. Click the icon in the **URLs** column.
3. In the pop-up, select the URL that you want to delete.
4. Click the deletion icon. The URL is removed immediately.
5. Click **OK**.
6. Save your model configuration.

Admissible Planned Events

To define planned events, you can use admissible planned events.

In the **Tracked Process** tab, click the drop-down of the **Content** view on the right and switch to the **Admissible Planned Events** view. All the admissible planned events are listed on the right side of this page.

By default, the content of **Admissible Planned Events** is displayed with a **User Model Planned Event Types** table. It contains the user-defined planned events in this tracked process. You can add/edit/delete these planned events.

Besides, in a standard tracked process for the standard model "gttft1", you can also see a **Standard Model Planned Event Types** table. It contains the standard planned events in this tracked process. You can only edit these planned events.

You can have at most 200 admissible planned events in a tracked process.

Create Admissible Planned Event

Context

Before doing this task, you must first define one or more event types in the **Event Type Pool** tab.

To create an admissible planned event, do the following:

Procedure

1. In the **Tracked Process** tab, click the drop-down of the **Content** view on the right and switch to the **Admissible Planned Events** view.
2. Click the **Add** button above the **User Model Planned Event Types** table. A pop-up dialog appears.
3. In the **Details** tab, fill in the following information:

Field	Explanation
Event Type*	Select an event type that you defined in the **Event Type Pool** tab. Note that • If the event type is already used by another admissible planned event in the same tracked process, this event type is not selectable and you are not able to select it as an admissible planned event again. • The events that inherit `GTTDelayedEvent` and `GTTOnTimeEvent` won't be displayed in the drop-down list.
Default Business Tolerance	It's used to calculate the earliest planned business timestamp and the latest planned business timestamp, if they are not provided in the event payload. Enter a number that is greater than or equal to 0. The default is 0.
Default Technical Tolerance	It's used to calculate the earliest planned technical timestamp and the latest planned technical timestamp, if they are not provided in the event payload. Enter a number that is greater than or equal to 0. The default is 0.
Periodic Overdue Detection*	It's used to define how often you want the system to check the status of overdue events. For example, if you enter 10 minutes, then the system will check the status of overdue events every 10 minutes. Enter a number that is greater than or equal to 5.
Max Overdue Detection*	It specifies the maximum times that you want the system to check the status of overdue events. For example, if you enter 3, the system will check the overdue events three times and then ignore them even if they are still overdue. Enter a number that is less than or equal to 5.
Affect Planned Event Status By	Select in the drop-down: